Sustainable Development Goals on air pollution, energy and health

WHO is the custodian agency of a handful of **SDG indicators** for Goals 3 (pertaining to good health and well-being), but also in Goals 2 (Zero hunger), 5 (Gender equality), 6 (Water and sanitation), 7 (Energy), 11 (Cities and communities) and 16 (Peace, justice and strong institutions) which highlights the importance of mainstreaming health across topics. Among them are **three air pollution-related indicators**:

SDG indicators

SDG 3.9.1 indicator: Mortality rate attributed to household and ambient air pollution[1]

SDG 7.1.2 indicator: Proportion of population with primary reliance on clean fuels and technologies[2]

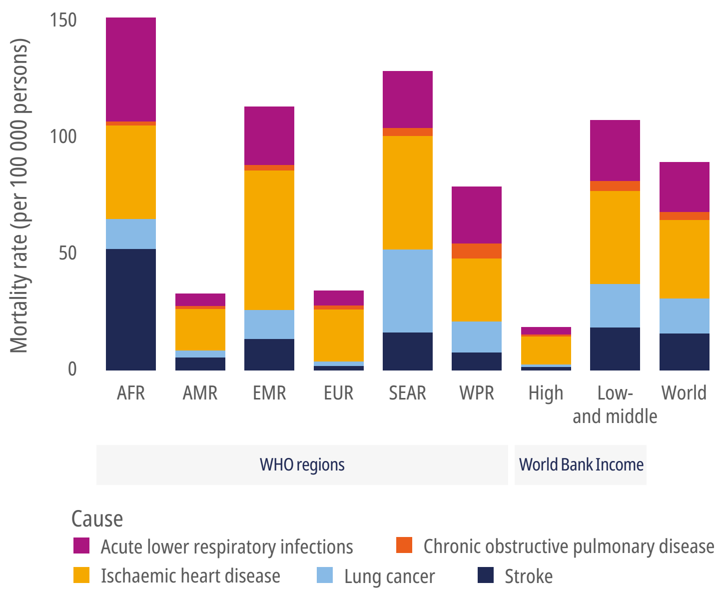
SDG 11.6.2: Annual levels of fine particulate matter in cities[3]

Air pollution related SDG indicators data is available at country level and for a wide range of regional and income levels.

World Health Organization

Key highlights

SDG 3.9.1 Mortality rate attributable to air pollution, by WHO regions, income levels and disease, 2021



6.6 million deaths

attributable to air pollution exposure

Around 90% of the burden statement:

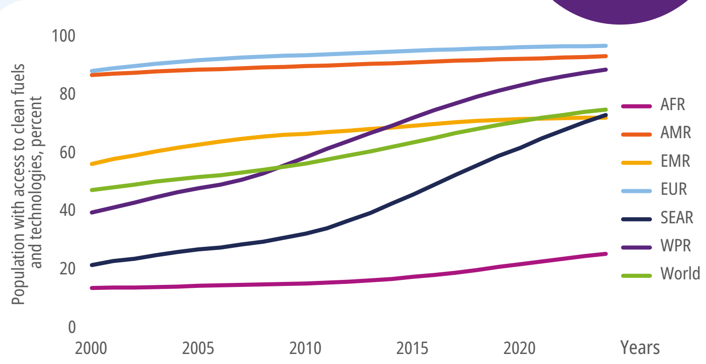
- is borne by low- and middle-income countries
- are due to NCDs in the European and Americas WHO regions

Acronyms: AFR – Africa; AMR – Ameria; EMR – Eastern Mediterranean; EUR – Europe; SEAR – South-East Asia; WPR – Western Pacific

SDG 7.1.2 Progress towards universal access to clean cooking, 2010–2024, by WHO regions

2 billion

still rely on polluting cooking

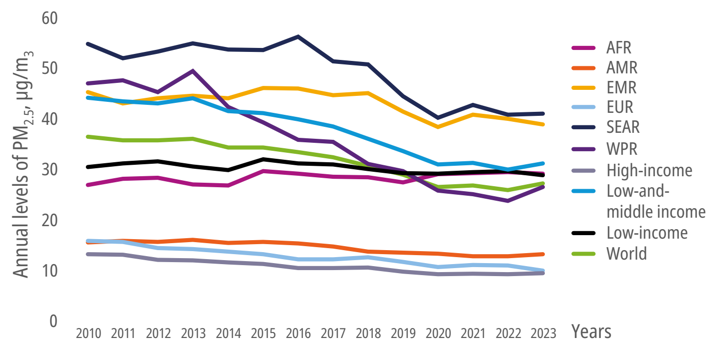


Progress has been made since 2010, with 75% of the global population having clean household energy access in 2024 but around **2 billion people** remain dependent on polluting fuels and technologies for cooking, with the remaining access gap increasingly concentrated in Sub-Saharan Africa and rural communities.

SDG 11.6.2 Trends for annual levels of $PM_{2.5}$ in urban areas[4], by WHO regions and income levels, 2010–2023

13x higher risk

of breathing unsafe air in low and middle-income countries



In 2023, the number of people exposed to air quality exceeding the least stringent interim target of 35 μg/m_3, proposed by WHO air quality guidelines was thirteen times higher in low- and middle-income countries than in high-income countries, affecting 6.5 billion people.

Globally, since 2010 annual levels of fine particulate matter ($PM_{2.5}$) have been decreasing until 2020 when air pollution levels have remained largely unchanged. Cities have also seen progress in air pollution, while rural areas of low income countries have not.

Acronyms: AFR – Africa; AMR – Ameria; EMR – Eastern Mediterranean; EUR – Europe; SEAR – South-East Asia; WPR – Western Pacific

WHO's role on monitoring and reporting

To emphasize multi-sectoral collaboration between health and other sectors

While the health sector is the recipient of the burden from environmental risks, it is rarely in the decision-making seat with little power or influence in other sectoral policies (energy, transport, agriculture, industry, waste). Tackling air pollution at its source is the most cost-efficient, sustainable, and equitable way to prevent disease and protect health in the long term, so any multisectoral actions should include the health voice.

To foster health impact assessments of air pollution while implementing sectoral policies

To make progress in air quality and protect the public, countries must understand their baseline exposures and associated health impacts. It is important to track air pollution and health impacts before, during and after any interventions to assess its effectiveness. Health impact and burden of disease assessments are cornerstone methods to provide decision-makers and stakeholders with comprehensive information about the health impacts and benefits of sectoral interventions.[5,6]

To monitor progress on achieving universal access to clean cooking by 2030

Household air pollution results from using polluting fuels and technologies for cooking, heating, and lighting. This does not only impact health directly but also has social, climate and environmental impacts and is a major source of ambient air pollution in many regions. Tracking clean household energy access is key to support implementation of the WHO indoor air quality guidelines: household fuel combustion and foster health and energy sectors engagement.

The knowledge of the relationship between air pollution and health impacts, and its quantification is fundamental to successful policies that protect our health and well-being.

WHO not only provides data and methods but also user-friendly tools to estimate adverse health risks and impacts of different air pollutants.

These tools empower the health sector to collaborate with other sectors.

How to strengthen country's capacity for health data?
WHO SCORE data collection tool

How to calculate air pollution related health impacts of sectoral interventions?
AirQ+ software tool

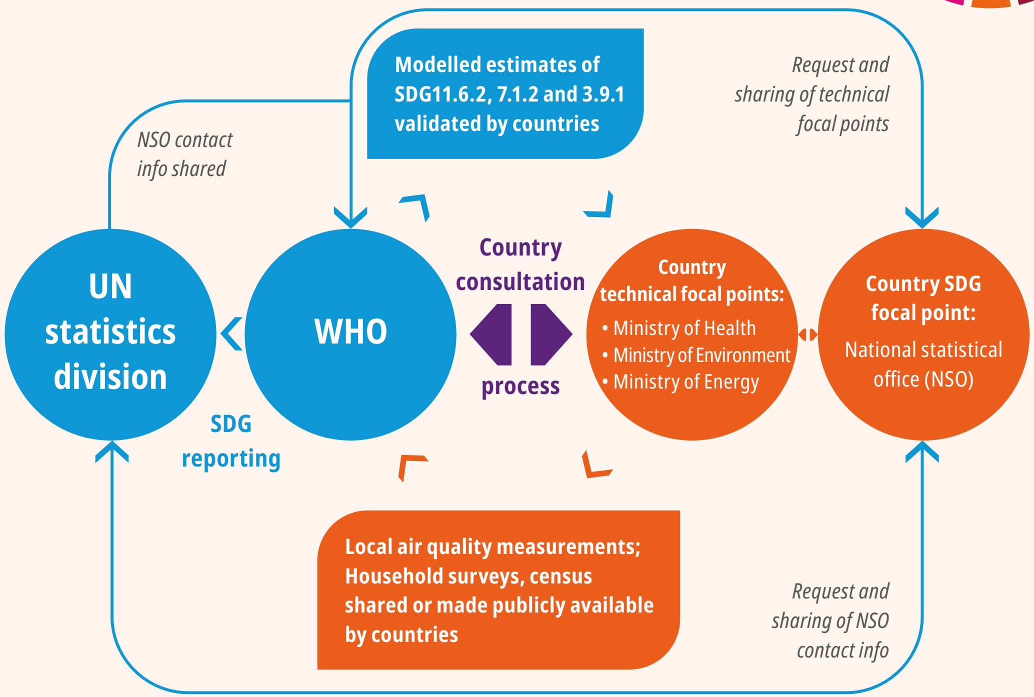
How to assess health impacts?

How to develop policy and action plans for clean household energy and use?
Clean Household Energy Solutions Toolkit (CHEST)

How does WHO collect, harmonize and validate data for global reporting?

Country consultation is a mandatory process during which WHO exchanges with countries on health-related data. During this formal process, WHO requests input data on air quality and household energy, forming a unique source of official data.

SDG reporting: how does it work?



SDG 3.9.1

Uses death registration data from the WHO Global Health Estimates (GHE), the backbone of the health estimates. To measure health risks from exposure, high-quality epidemiological evidence, matching the same strength of evidence informing the WHO air quality guidelines.

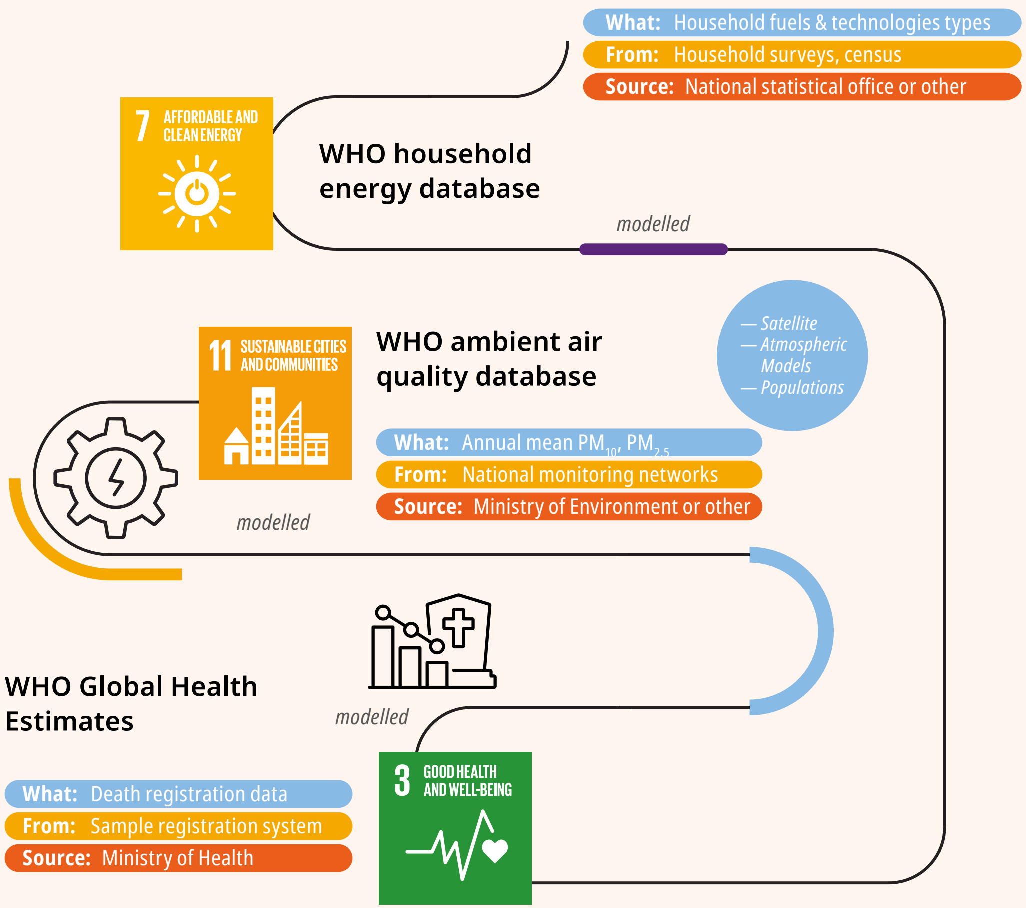
SDG 7.1.2

Household surveys, census data for type of household fuel use are required to inform the modelling exercise; compiled in the WHO household energy databases.

SDG 11.6.2

A global satellite product, combining multiple observations and atmospheric models, is calibrated using national measurements from WHO air quality database and population data from the Global Human Settlement Layer.

How do country data feed into the SDGs?



*The comparability of **SDG data** will be fundamental to track countries' progress towards achieving the WHO's proposed*

50% reduction

in the fraction of mortality attributed to anthropogenic air pollution

from 2015 to 2040

This data is the beginning; its journey toward better health is up to you

While countries may have their own air pollution exposure and burden and/or SDG estimates, this data provides a comparative dataset that can be used as basis in their absence. Data is the foundation for many actions such as conducting health impact assessments, monitoring progress of policy action, leveraging health sector for engaging in advocacy and bolstering academic research.

Countries have pledged to achieve the 50% reduction target proposed in the WHO's updated road map for air pollution[7]. All these efforts aim to support WHO's target of a 50% reduction in air pollution-related mortality by 2040, while contributing to the SDG target of universal access to clean cooking by 2030 and the global roadmap ambition to eliminate cooking poverty.

While WHO's updated road map target focuses on anthropogenic air pollution, it is important to highlight that natural sources of air pollution such as wildfire smoke[8] and sand and dust storms[9] which are transboundary by nature and depend on meteorology, topography, have increased in the last years as a result of human activity.

SDG air pollution estimates

STRENGTHS	LIMITATIONS
Transparent process	Conservative estimates, not necessarily reflecting latest scientific findings
Country official data used	Longer process time to ensure country involvement
Standard UN population datasets used to allow comparisons across countries and years	Gaps in data, especially from data-poor countries
Consistent methodology easy to communicate	

Besides countries, what stakeholders are important in developing and using this data?

Data development: the scientific community, such as environmental epidemiologists, air pollution exposure scientists, atmospheric scientists, and statisticians are involved through various fora and WHO-convened expert groups, such as the WHO Global Air Pollution and Health – Technical Advisory Group and Scientific Advisory Group and the World Meteorological Organization (WMO) Global Atmospheric Watch Programme.

Ongoing collaborations with the Universities of Manchester, United Kingdom & George Mason University, USA (SDG 11.6.2 modelling) and Glasgow, United Kingdom (SDG 7.1.2 modelling).

Data users: UN agencies, development partners and financing institutions: UNEP, UNICEF, UN-Habitat, UN DESA, UN Energy, World Bank, as well as civil society and advocacy groups.

How to access and use the data

WHO database on air quality,

household energy database

air quality standards

WHO's Global Health Observatory

UN SDG website

WHO Science and Policy Summaries

WHS 2026

SDG 7 Tracking

WHO's Air pollution and health training toolkit

Footnotes

1 Age-standardized mortality rate attributed to ambient and household air pollution (per 100 000 population). Data are available from 2010 to 2021 and include deaths, disability-adjusted life-years (DALYs), years of life lost (YLLs) and years lived with disability (YLDs) attributed to air pollution by disease (acute lower respiratory infections, chronic obstructive pulmonary disease, ischaemic heart disease, lung cancer and stroke), sex and age

2 Proportion of population with primary reliance on clean fuels and technologies (percentage of population) available from 2000 to 2024 across two geographic areas: urban and rural areas

3 Population-weighted annual levels of fine particulate matter in cities (micrograms per cubic metre) available from 2010 to 2023 across three geographic areas: cities, towns and rural areas

4 Urban areas is a spatial aggregation of cities and towns as defined by the Degree of Urbanization

5 World Health Organization. Health science and policy summaries [website]. Geneva: World Health Organization; 2024 [accessed 23 June 2024]. https://www.who.int/teams/environment-climate-change-and-health/air-quality-energy-and-health/health-impacts/health-science-and-policy-summaries

6 World Health Organization. AirQ+ software tool: software for health impact assessment of air pollution [software]. Geneva: World Health Organization; 2023 [accessed 23 June 2024]. https://www.who.int/tools/airq

7 https://apps.who.int/gb/ebwha/pdf_files/EB156/B156_24-en.pdf

8 World Health Organization. Wildfires [website]. Geneva: World Health Organization; 2024 [accessed 23 June 2024].
https://www.who.int/health-topics/wildfires

9 World Health Organization. WHO science and policy summaries: Understanding the health impacts of sand and dust storms. Geneva: World Health Organization; 2024 [accessed 23 June 2024]. https://iris.who.int/handle/10665/382119

World Health Organization